

Group AN — North America + Europe

May 2026

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1. Data Loading and Investment Sets

1.1 Data Sources

The project uses processed investment set files (CSV format) for each year from 2013 to 2025. Each file, named `investment_set_{year}.csv`, contains one row per firm and the following key variables:

Variable	Description
ISIN	Firm identifier
NAME, Country, Region	Firm identifiers and geographic classification
MV_Y	End-of-year market capitalisation (million USD)
CO2_S1, CO2_S2	Scope 1 and Scope 2 CO emissions (tonnes)
REV	Annual revenues (thousands USD)
CI	Carbon intensity = $(\text{CO2_S1} + \text{CO2_S2}) / (\text{REV} / 1000)$ (tCOe per million USD revenue)
Monthly return columns	120 monthly total returns (10-year estimation window ending in year Y)

Carbon data covers 2002–2024. Due to limited coverage before 2010, the allocation exercise begins at end of 2013 (portfolio first invested in 2014).

1.2 Investment Set Construction

For each year $Y = 2013, \dots, 2024$, the investment set is defined as firms satisfying all of the following criteria:

- Belong to the assigned region.
- Have sufficient return history: at least 3 years (36 months) of non-missing monthly returns over the 10-year window ending at end of year Y.
- Have carbon data (CO and revenues) available at end of year Y.
- Do not have stale prices: firms whose monthly returns are zero for more than 50% of the estimation window are excluded.

The investment decision is made at end of year Y for portfolio implementation over year Y+1. The covariance matrix and expected returns are estimated from the 10-year window of monthly returns available at end of Y.

1.3 Full-Available-Data Variant

A second version of each investment set, labelled `is_{year}_fa`, is constructed by dropping any firms with at least one missing monthly return over the 10-year window. This stricter filter is used for the full-available-data value-weighted portfolio and as the benchmark in the Ledoit-Wolf tracking-error and net-zero optimizations.

2. Value-Weighted Portfolios

2.1 Standard Value-Weighted Portfolio

At the end of each year Y, portfolio weights are set proportional to each firm’s market capitalisation within the investment set:

$$w_{\{i,Y\}} = MV_{\{i,Y\}} / \sum_j MV_{\{j,Y\}}$$

These end-of-year weights are stored in the column `MV_Y_weight` and serve as the January weight for year $Y+1$. However, rather than holding these weights fixed throughout the year, the portfolio is reweighted each month using the actual end-of-month market capitalisations from `MV_M_filtered.csv`. Concretely, for February through December of year $Y+1$, the weight of firm i at the start of month t is:

$$w_{\{i,t\}} = MV_{\{i,t-1\}} / \sum_j MV_{\{j,t-1\}} \text{ for } t = Feb(Y+1), \dots, Dec(Y+1)$$

where $MV_{\{i,t-1\}}$ is the market capitalisation of firm i at the end of the preceding month, taken from `MV_M_filtered.csv`. Monthly caps are matched to return columns by nearest available date. If a monthly cap is unavailable, the end-of-year market cap (MV_Y) from the prior investment set is used as a fallback. Weights are renormalized within the investment-set universe so they sum to 1 each month. The monthly portfolio return is then $R_{\{p,t\}} = \sum_i w_{\{i,t\}} \times R_{\{i,t\}}$, where missing returns are replaced by zero to avoid survivorship bias (firms that delist or have gaps are treated as contributing 0 return rather than being dropped from the denominator).

2.2 Full-Available-Data Value-Weighted Portfolio

The same monthly market cap reweighting methodology is applied to the `is_{year}_fa` subset (firms with complete 10-year return histories). After removing firms with any NaN in their return series, the end-of-year `MV_Y_weight` values no longer sum to 1 over the reduced universe; they are therefore explicitly renormalized so that $\sum_i w_{\{i,Y\}}^{fa} = 1$. Monthly reweighting from February onwards then uses the same `MV_M_filtered.csv` caps restricted to the `fa` universe, renormalized within that universe. This portfolio is stored as `MV_Y_weight_fa` and serves as the benchmark for tracking-error and net-zero optimizations.

3. Minimum-Variance Portfolio Optimization

3.1 Covariance Matrix Estimation

For each year Y , two covariance matrices are computed from the 10-year window of monthly returns:

Sample Covariance (SLSQP method): The standard sample covariance matrix $_Y = (1/_k) \sum_{k=0}^{_k-1} (R_{\{t-k\}} - _Y)(R_{\{t-k\}} - _Y)'$ where $_k = 120$ months. Computed using `pandas.cov()` on the transposed return matrix.

Ledoit-Wolf Shrinkage (CVXPY method): Before applying Ledoit-Wolf, firms with any NaN in their return series are removed. The remaining returns are passed to `scikit-learn`'s `LedoitWolf` estimator, which produces a positive-semidefinite covariance matrix $_Y^{LW}$ by optimally shrinking the sample covariance towards a scaled identity. The resulting subset of firms is stored as `returns_{year}_lw`.

3.2 SLSQP Optimization (Sample Covariance)

The minimum-variance weights for year Y are obtained by solving:

$$\min_{_i} _i' _Y$$

$$s.t. _i _i = 1, _i \geq 0 \text{ for all } i$$

Solved with `scipy.optimize.minimize` using the SLSQP method. A numerical scaling factor of 10,000 is applied to $_Y$ before optimization to improve convergence, and the resulting weights are verified to be identical to those from the unscaled problem. The initial guess is the equal-weight vector.

3.3 CVXPY / OSQP Optimization (Ledoit-Wolf)

The same quadratic program is solved with the CVXPY library using the OSQP solver on the Ledoit-Wolf covariance matrix $_Y^{\wedge}\{LW\}$:

$\min_w \ w' _Y^{\wedge}\{LW\} \ w \ (CVXPY \ quad_form)$

$s.t. \ _i \ w_i = 1, \ w_i \geq 0$

A scaling factor of 10,000 is also applied for numerical stability. Small negative weights produced by solver numerical noise are clipped to zero and weights are renormalized to sum to 1.

3.4 Out-of-Sample Returns

For each year $Y = 2013, \dots, 2024$, the weights $_Y$ computed at end of Y are applied to the actual monthly returns of year $Y+1$. The portfolio is rebalanced annually; within the year, weights drift with prices (dynamic weight update). This procedure produces a 144-month out-of-sample return time series from January 2014 to December 2025.

4. Carbon Metrics

4.1 Carbon Intensity (CI)

Carbon intensity measures CO emissions relative to revenues. For each firm i in year Y :

$$CI_i\{Y\} = (CO2_S1_i\{Y\} + CO2_S2_i\{Y\}) / (REV_i\{Y\} / 1000)$$

The division by 1000 converts revenues from thousands to millions of USD, giving CI in tonnes of CO equivalent per million USD of revenue (tCOe / M\$rev).

4.2 Weighted Average Carbon Intensity (WACI)

The portfolio-level WACI aggregates firm-level carbon intensity using portfolio weights:

$$WACI_Y^{\wedge}\{p\} = _i _i\{Y\} \times CI_i\{Y\}$$

Computed annually for each portfolio using the weights determined at end of year Y .

4.3 Carbon Footprint (CF)

The carbon footprint of the portfolio measures the CO emissions attributed to the investor per million USD invested:

$$CF_Y^{\wedge}\{p\} = _i _i\{Y\} \times E_i\{Y\} / Cap_i\{Y\}$$

where $E_i\{Y\} = CO2_S1_i\{Y\} + CO2_S2_i\{Y\}$ (total emissions in tonnes), $Cap_i\{Y\} = MV_Y$ (market capitalisation in million USD), and $_i\{Y\}$ is the portfolio weight. The result is in tonnes of CO per million USD invested (tCOe / M\$inv). Portfolio wealth cancels out of this formula, so no wealth evolution is required.

5. CF-Constrained Minimum-Variance Portfolio (Part 3.2)

A carbon footprint constraint is added to the minimum-variance problem to force the portfolio CF to be no more than 50% of the unconstrained minimum-variance portfolio CF in the same year:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \sigma^2(\mathbf{w}) \\ \text{s.t.} \quad & CF_Y(\mathbf{w}) \leq 0.5 \times CF_Y(mv) \\ & \sum_i w_i = 1, w_i \geq 0 \end{aligned}$$

where $CF_Y(mv)$ is the carbon footprint of the unconstrained minimum-variance portfolio computed for year Y . The constraint is re-evaluated each year so that the 50% reduction target is relative to the current unconstrained portfolio, not a fixed baseline.

Two implementations are produced:

- SLSQP (sample covariance): same solver as Section 3.2, warm-started from equal weights.
- CVXPY/OSQP (Ledoit-Wolf): same solver as Section 3.3, with the CF constraint expressed as a linear inequality $\mathbf{e}^T \mathbf{w} \leq \text{target}$ where $\mathbf{e}_i = E_{i,Y}/Cap_{i,Y}$.

6. Tracking Error Minimization (Part 3.3)

This strategy minimizes the tracking error relative to the value-weighted benchmark while constraining the portfolio carbon footprint to 50% of the VW benchmark CF. It targets a passive investor seeking to reduce carbon emissions with minimal deviation from the market-cap portfolio.

The optimization problem is:

$$\begin{aligned} \min_{\mathbf{w}} \quad & TE_Y^2 = (\mathbf{w} - \mathbf{w}^{vw})^T \Sigma_Y (\mathbf{w} - \mathbf{w}^{vw}) \\ \text{s.t.} \quad & CF_Y(\mathbf{w}) \leq 0.5 \times CF_Y(vw) \\ & \sum_i w_i = 1, w_i \geq 0 \end{aligned}$$

Minimizing TE^2 is equivalent to minimizing TE since the square-root function is strictly increasing over non-negative values and is numerically preferable near zero.

Two implementations:

- SLSQP: uses the sample covariance matrix and MV_Y_weight as the VW benchmark. Warm-started from the VW weights to ensure feasibility.
- CVXPY/OSQP with Ledoit-Wolf: uses Σ_Y^{LW} and $MV_Y_weight_fa$ (the full-available-data VW benchmark). The VW_fa weights are reindexed to the LW subset of firms and renormalized to sum to 1 before use in the objective. The CF reference is computed over the full investment set using $MV_Y_weight_fa$.

7. Net Zero Portfolio (Part 4.1)

The net zero strategy adopts the same tracking-error minimization framework as Section 6, but replaces the static 50% CF reduction constraint with a time-varying constraint that enforces a compounding 10% annual reduction in the portfolio carbon footprint from a fixed 2013 baseline:

$$CF_Y^{\{(p)\}} (1 -)^{\{YY_0+1\}} \times CF_Y^{\{(vw)\}}$$

where $\alpha = 0.10$, $Y_0 = 2013$, and $CF_Y^{\{(vw)\}}$ is the carbon footprint of the value-weighted portfolio in 2013 (computed once as a fixed reference). This gives the following annual carbon budgets:

Year Y	CF budget as % of 2013 VW CF
2013 (YY+1 = 1)	90.0% (0.9 ¹)
2014 (exponent = 2)	81.0% (0.9 ²)
2016 (exponent = 4)	65.6% (0.9)
2019 (exponent = 7)	47.8% (0.9)
2022 (exponent = 10)	34.9% (0.9 ¹)
2024 (exponent = 12)	28.2% (0.9 ¹²)

The net-zero strategy is implemented using the CVXPY/Ledoit-Wolf solver only. The SLSQP implementation was removed as the LW formulation provides a more robust and numerically stable solution for this constrained problem.

- CVXPY/LW: CF reference = 2013 VW_fa CF using MV_Y_weight_fa weights computed over the fa universe. CVXPY weights are stored with `.flatten()` to ensure a 1-D array regardless of solver output shape.
- The `nz_weight_slsqp` column is no longer populated in the investment set CSVs; only `nz_weight_lw` is stored.

CVXPY weights are stored with `.flatten()` to ensure a 1-D array regardless of solver output shape, avoiding downstream dtype errors.

8. Portfolio Weight Storage and CSV Management

All portfolio weights are collected into a unified per-year investment set DataFrame (`inv_set_{year}`) and persisted to CSV to allow inter-cell continuity without re-running the full optimization pipeline. The final column structure is:

Column Index	Contents
0–4	ISIN, NAME, Country, Region, MV_Y
5	MV_Y_weight (value-weighted)
6	MV_Y_weight_fa (full-available-data VW)
7	min_var_weight_slsqp (unconstrained MV, SLSQP)
8	min_var_weight_lw (unconstrained MV, LW)
9	min_var_weight_cf_slsqp (CF-constrained MV, SLSQP)
10	min_var_weight_cf_lw (CF-constrained MV, LW)
11	te_weight_slsqp (tracking error, SLSQP)
12	te_weight_lw (tracking error, LW)
13	nz_weight_slsqp (net zero, SLSQP — removed, column not populated)
14	nz_weight_lw (net zero, LW)
15+	CO2_S1, CO2_S2, REV, CI, and monthly return columns

When adding LW-specific weights (which are only defined for the NaN-free LW subset), a left-merge on ISIN is performed so that firms outside the LW universe receive a weight of 0. Stale columns from previous runs are dropped before merging to avoid pandas 2.0+ MergeError on duplicate column names.

9. Out-of-Sample Return Computation

9.1 Return Dataframe Format

Returns are stored in a wide format using the `create_returns_dataframe` helper: a DataFrame with years (2014–2025) as the row index and months (Jan–Dec) as the twelve columns. A monthly time series is obtained by calling `.stack()` on this DataFrame, producing a Series indexed by (year, month) tuples.

9.2 Value-Weighted Portfolio Returns

For the value-weighted portfolios (both full investment set and fa subset), the return for each month of year $Y+1$ is computed using the appropriate monthly weight column. Return columns and weight columns are paired by sorting both independently by date, guaranteeing correct alignment regardless of column order in the DataFrame. The steps are:

- Build a per-year DataFrame containing `MV_Y_weight` (January weight) and `MV_M_weight_{date}` columns (February through November weights) from actual monthly market caps, plus the 12 monthly return columns from `is_{year}`.
- For each month t , multiply the corresponding weight column by the return column (with missing returns filled as zero) and sum across all firms to obtain $R_{\{p,t\}}$.
- December uses the November end-of-month cap weights (the 11th `MV_M` column), which together with `MV_Y_weight` gives exactly 12 weight-return pairings per year.

9.3 Minimum-Variance, CF-Constrained, Tracking-Error, and Net-Zero Portfolio Returns

All optimized portfolios (SLSQP, LW, CF-constrained, tracking-error, and net-zero) use a buy-and-hold weight drift approach rather than fixed monthly weights. Weights are set once at end of year Y and then allowed to evolve naturally as asset prices move, exactly as they would for an investor who invests at the start of the year and does not rebalance intra-year. The update rule for each month t within year $Y+1$ is:

- $w_{\{i,t\}} = w_{\{i,t-1\}} \times (1 + R_{\{i,t-1\}}) / (1 + R_{\{p,t-1\}})$
- An asset whose return exceeds the portfolio return this month will have a higher weight next month; an asset that underperforms will shrink in share. The division by $(1 + R_{\{p,t-1\}})$ ensures that the updated weights still sum to 1 at all times. Missing returns are filled as zero so that delisted or temporarily absent firms contribute 0 to portfolio return without distorting the weight denominator (no survivorship bias).
- This procedure is applied identically to all optimized strategies: SLSQP min-variance, LW min-variance, CF-constrained (SLSQP and LW), tracking-error (SLSQP and LW), and net-zero

(LW only). The result is a 12-element monthly return series per year, stacked across years to form the full 144-month out-of-sample series (January 2014 to December 2025).

9.4 Performance Statistics

For each portfolio, the following statistics are computed over the full 144-month period (January 2014 – December 2025):

Statistic	Formula
Annualized Average Return	$(1 + \text{mean}(r_monthly))^{12} - 1$
Annualized Volatility	$\text{std}(r_monthly) \times 12$
Sharpe Ratio	Annualized Return / Annualized Volatility
Annualized Cumulative Return	$(1 + \text{cum_return})^{\{1/12\}} - 1$
Min/Max Monthly Return	min and max of $r_monthly$

10. Carbon Footprint and WACI Evolution

For each portfolio and each year $Y = 2013, \dots, 2024$, the annual CF and WACI are computed from the weights determined at end of Y and the carbon data for year Y . For LW portfolios, the weight vector is reindexed from the LW-subset order to the full `inv_set` order using a ISIN-based reindex, and emission intensities are computed over the matching LW subset.

Results are collected into `cf_waci_*_df` DataFrames indexed by year, enabling line plots that show the evolution of both metrics across the sample period for all portfolio variants.

11. Top-10 Carbon Intensive Firms Analysis

For each year and each portfolio, the 10 firms with the highest carbon intensity (CI) are identified from the investment set. Their portfolio weights are displayed as horizontal bar charts in a 4×3 grid (one subplot per year, 2013–2024). Each portfolio family uses a consistent colour scheme:

Portfolio Family	Bar Colour
Unconstrained MV (SLSQP and LW)	Steelblue
CF-Constrained MV (SLSQP and LW)	Tomato
Tracking Error (SLSQP and LW)	Medium Sea Green
Net Zero (LW only)	Dark Orchid

These plots allow visual inspection of how much weight each strategy allocates to the most carbon-intensive names, and how this evolves as constraints tighten.

12. Decomposition of CF and WACI Changes

A counterfactual decomposition separates the observed change in CF and WACI from the 2013 level into two effects, following a Laspeyres-type framework:

12.1 Emission Effect (frozen weights)

The portfolio weights are held fixed at their 2013 values while firm-level emissions and carbon intensity are allowed to evolve. This isolates the contribution of firm-level decarbonization (or deterioration) to the portfolio-level carbon metric, independently of reweighting decisions.

$$CF_Y^{\{emission\}} = \sum_i (w_{\{i,2013\}} / \sum_j w_{\{j,2013\}}) \times E_{\{i,Y\}} / Cap_{\{i,Y\}}$$

12.2 Composition Effect (frozen emissions)

The firm-level emissions and carbon intensities are held fixed at 2013 values while the portfolio weights are allowed to vary. This isolates the contribution of active portfolio reweighting (tilting towards lower-carbon firms) to the portfolio-level carbon metric.

$$CF_Y^{\{composition\}} = \sum_i w_{\{i,Y\}} \times E_{\{i,2013\}} / Cap_{\{i,2013\}}$$

The same framework is applied to WACI, substituting CI for E/Cap. Each decomposition figure contains two panels (WACI left, CF right), and is produced for each portfolio variant. Only firms present in both the 2013 and year-Y investment sets (inner merge on ISIN) contribute to the decomposition.

13. Summary of Portfolio Families

Portfolio	Description
VW	Value-weighted, standard MV_Y weights
VW (full available data)	Value-weighted, NaN-free firms only, renormalized
MV (SLSQP)	Min-variance, sample covariance, long-only
MV (LW)	Min-variance, Ledoit-Wolf shrinkage, long-only
MV CF-Constr. (SLSQP)	Min-variance + CF 50% of MV CF (sample cov.)
MV CF-Constr. (LW)	Min-variance + CF 50% of MV CF (LW)
TE (SLSQP)	Tracking-error min. + CF 50% of VW CF (sample cov.)
TE (LW)	Tracking-error min. + CF 50% of VW_fa CF (LW)
NZ (LW only)	TE min. + CF $(0.9)^{Y-2013} \times CF_{\{2013\}}^{\{VW_fa\}}$ (LW, CVXPY/OSQP)
NZ — removed	SLSQP net-zero implementation removed; LW-only net-zero listed above

Results

Why the drastic decrease in CF and WACI?

The issue is not a computation error but rather what the investment set actually contains in each year. The investment set for 2013 is restricted to firms that had carbon data available by end-2013 — mostly large, established companies in energy, materials, and industrials, which are precisely the most carbon-intensive sectors. In subsequent years, as carbon reporting improves and more firms start disclosing emissions, many new companies enter the investment set. These are predominantly tech, healthcare, and services companies — sectors with very low carbon intensity and, crucially, very high market capitalizations.